

Question Number	Answer	Additional Guidance	Mark
3a	There is no (significant) difference between the number of pupae from the untreated (A) and treated culture (B) medium	Ignore eggs hatched	1 exp

Question Number	Answer	Additional Guidance		Mark
3b	- suitable table format with data correct column headings (1) - means correctly calculated (1)	Number of pupae		
		A/untreated	B/treated	
		72	45	
		68	56	
		81	39	
		56	40	
		43	29	
		52	38	
		60	35	
		64	46	
		Mean 62	Mean 41	

Question Number	Answer	Additional Guidance	Mark
3c	- bar graph with linear scale and labels (1) - means plotted correctly (1) - range bars plotted correctly (1)	mean number of pupae/culture or treatment/ A B. y axis must start at zero ALLOW ECF for incorrect means from 3bi Range bars 81-43 56-29	3 exp

Question Number	Answer	Additional Guidance	Mark
3di	- correct substitution of given $(S_A)^2$ and $(S_B)^2$ (1) - correct answer (1)	ECF allow use of incorrect means Correct square root reduces to 5.109 $t = 4.11$ (4.1) Correct answer only gains 2 marks accept additional decimal places	2 exp

Question Number	Answer	Additional Guidance	Mark
3d ii	- the calculated value of t (4.11) is more than the critical value 2.14 (1) - therefore reject the null hypothesis there is a difference between the treated and untreated groups (1)	Allow ECF from 3di if calculated value is less than stated critical value Accept critical value of 2.98	2 exp

Question Number	Answer	Additional Guidance	Mark
3e	An answer that includes two of the following points: • use other masses of pyrethrum (1) • leave adults to lay eggs for more than 24 hours (1) • Leave eggs for more than 5 days (to see if more hatch/pupate) (1)	ignore repeat the expt/amount accept concentration/volume accept more time accept more time	2 exp
Question Number	Answer	Additional Guidance	Mark
3f	An answer that includes two of the following points: • Pyrethrum/concentration/ mass may have no effect on the flies/flies resistant (1) • Pyrethrum may be washed/blown off fruits (1) • pyrethrum may not reach eggs (1) • fruit damaged by {being eaten/pests/weather} (1)	accept flies become resistant Fruit skin might prevent pyrethrum entering fruit	2 exp

(Total for question 3 = 14 marks)

